Radiation Physics and Chemistry 245 (2026) 113893

Contents lists available at ScienceDirect

Radiation Physics and Chemistry

journal homepage: www.elsevier.com/locate/radphyschem

Impedance spectra of rhodium-doped silicon irradiated with alpha particles

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ARTICLE INFO

Keywords:
Silicon
Rhodium
Doping
Diffusion
Alpha particles
Impedance spectroscopy

ABSTRACT

In this work, the effect of irradiation with α-particles at fluences of 5×10^{14} and 1×10^{15} cm^{-2} on the electrophysical properties of rhodium-doped n-type silicon is investigated by the method of impedance spectroscopy as a function of the diffusion temperature. It is established that irradiation of undoped n-Si leads to a significant degradation of its electrical characteristics, accompanied by a sharp increase in resistivity due to the formation of radiation-induced defects and deep levels in the forbidden energy gap. It is shown that with increasing irradiation fluence, the resistance of the samples increases by more than four times compared to the initial state. A comparative analysis of the results indicates a substantial influence of both the irradiation dose and the rhodium diffusion temperature on the electrophysical parameters of n-Si. At the same time, rhodium doping significantly reduces the degree of radiation-induced degradation of the material, especially at a diffusion temperature of 1200 °C, which indicates its stabilizing effect on the defect structure of silicon.

1. Introduction

At present, silicon remains the primary material of modern micro- and nanoelectronics and is also widely used in power electronics, solar energy systems, and radiation-hardened semiconductor devices. Under operating conditions in outer space, nuclear energy systems, and accelerator technology, semiconductor structures are inevitably exposed to various types of ionizing radiation, including α-particles, protons, electrons, and γ-quanta. Such exposure leads to degradation of the electrophysical parameters of devices, which significantly limits their service lifetime and operational reliability [Tolasa, 2025; Utamuradova et al., 2025a].

It is well known that irradiation of silicon with high-energy particles is accompanied by the formation of radiation-induced defects, such as vacancies, interstitial atoms, and their complexes, which introduce deep levels into the forbidden energy gap. These levels efficiently trap charge carriers and act as recombination centers, resulting in a reduction of the free carrier concentration and deterioration of the conductive properties of the material. As a consequence, parameters such as resistivity and bulk resistance increase substantially, and the impedance characteristics of semiconductor structures are also modified [Oblakowska-Mucha et al., 2017; Utamuradova et al., 2023a].

One of the promising approaches to improving the radiation hardness of silicon is its intentional doping with various impurities capable of interacting with radiation defects and forming electrically less active complexes. In this context, transition metals are of particular interest, as they can significantly modify the defect subsystem of silicon and affect the processes of defect generation and annihilation under irradiation. However, the influence of specific impurities and the conditions of their incorporation into the crystal have not yet been sufficiently studied [Aboy et al., 2014; Maanzo et al., 2025].

Rhodium is one of such impurities that is potentially capable of modifying the defect–impurity structure of silicon and, consequently, its electrophysical and radiation-related characteristics. Depending on the diffusion doping temperature, rhodium may be distributed differently throughout the crystal volume and form various complexes with intrinsic and radiation-induced defects. This suggests a strong dependence of the radiation resistance of silicon on the doping temperature and the concentration of the introduced impurity [Istratov et al., 2000; Sourov et al., 2017].

In addition to the above considerations, the electrical properties of semiconducting crystals are strongly governed by defect-related

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https://doi.org/10.1016/j.radphyschem.2026.113893
Received 14 January 2026; Received in revised form 31 March 2026; Accepted 4 April 2026
Available online 4 April 2026

conductivity mechanisms and their frequency-dependent dielectric response. Radiation-induced defects introduce localized states within the band gap, which contribute to hopping conduction, polarization processes, and relaxation phenomena observable in the alternating current regime. As a result, the dielectric permittivity and loss tangent of semiconductor materials exhibit a pronounced dependence on frequency, reflecting the dynamics of charge carrier trapping and detrapping at defect sites. Such behavior has been extensively studied in various semiconductor systems and provides valuable insight into the role of defect structures in determining the overall electrical performance of irradiated materials. In this regard, the analysis of frequency-dependent electrical characteristics is essential for a comprehensive understanding of defect-induced modifications in semiconductors [Abdullayev et al., 2024].

From a structural point of view, the incorporation of transition metal impurities such as rhodium into silicon can significantly influence the stability of the crystal lattice under extreme conditions. Previous studies indicate that rhodium atoms are capable of enhancing interatomic bonding within the silicon matrix, thereby improving the structural integrity of the material. In particular, rhodium has been shown to suppress radiation-induced amorphization under high-energy γ-irradiation, effectively preserving the crystalline order. This effect is often associated with the ability of rhodium to act as a stabilizing center or "anchor" within the lattice, limiting the propagation of defect clusters and preventing large-scale lattice degradation at high radiation doses. Consequently, the introduction of rhodium not only modifies the defect subsystem but also contributes to improved radiation hardness and long-term reliability of silicon-based devices operating in harsh radiation environments [Utamuradova et al., 2025b].

Electrochemical impedance spectroscopy (EIS) is a powerful and informative method for investigating the electrical properties of semiconductor materials and structures, allowing the contributions of bulk and interfacial processes to be separated, as well as providing information on charge transport mechanisms and relaxation processes. Analysis of Nyquist plots makes it possible to quantitatively determine changes in bulk resistance and capacitive characteristics caused by both doping and radiation exposure [Barsoukov and Macdonald, 2005; Vladikova et al., 2011].

In the present work, the effect of irradiation with α-particles of 2 MeV energy at fluences of 5×10^{14} and 1×10^{15} cm^{-2} on the impedance characteristics of n-type silicon doped with rhodium by diffusion at temperatures of 1100 °C and 1200 °C is investigated. The main objective of this study is to establish the role of the rhodium doping temperature in the formation of radiation hardness of n-Si and to identify the mechanisms responsible for stabilizing its electrophysical parameters under intense radiation exposure.

2. Experimental part

Single-crystal n-type silicon wafers with a resistivity of approximately 40 Ω cm (KEF-40 grade) were used as the objects of investigation. The initial samples were cut from silicon ingots grown by the Czochralski method, which ensures high structural perfection of the material and good reproducibility of its electrophysical parameters. Prior to technological processing, all wafers were subjected to standard chemical cleaning to remove surface contaminants and the native oxide layer.

Rhodium doping of silicon was carried out by the diffusion method. For this purpose, a rhodium layer was preliminarily deposited onto the surface of the silicon wafers, after which the samples were placed into evacuated and sealed quartz ampoules. Thermal diffusion was performed at temperatures in the range of 1100–1200 °C for 3–5 h. The choice of these temperatures and processing times was dictated by the necessity to ensure sufficient impurity penetration depth and its uniform distribution in the near-surface region of silicon. A temperature of 1100 °C corresponds to the initial stage of activation of diffusion processes, while 1200 °C ensures deeper penetration of the impurity and intensive interaction with the native defects of silicon. After completion of the thermal treatment, the samples were cooled to room temperature under controlled conditions.

The initial and doped samples were then subjected to irradiation with α-particles with an energy of 2 MeV. The irradiation was performed at two fluence values: 5×10^{14} cm^{-2} and 1×10^{15} cm^{-2}. The irradiation experiments were carried out using the EG-5 electrostatic accelerator at the Laboratory of Neutron Physics of the Joint Institute for Nuclear Research (LNP JINR). The choice of these doses was motivated by the need to simulate moderate and severe radiation exposure of the silicon crystal structure and to analyze the evolution of its electrophysical properties.

Measurements of the electrical characteristics of the samples were performed by impedance spectroscopy at room temperature ($T \approx 300$ K). An Ellins P-45X impedance meter was used in a two-electrode measurement cell. The frequency range of the measurements was from 1 Hz to 0.5 MHz, and the amplitude of the applied sinusoidal signal was set to 10 mV, which ensured a linear response regime of the investigated structures and excluded the influence of nonlinear effects. The obtained data were processed in the form of Nyquist plots, from which the main parameters of the equivalent electrical circuit of the samples, primarily their bulk resistance, were determined.

A Tescan Vega3 scanning electron microscope (SEM) was used to study the surface morphology and to assess the structural state of the samples. The micrographs made it possible to monitor the surface quality after diffusion doping and irradiation, as well as to reveal possible signs of radiation-induced modifications in the near-surface region.

The elemental composition of the initial and doped samples was determined by energy-dispersive X-ray spectroscopy (EDS) implemented in the electron microscope. This method made it possible to confirm the presence of rhodium in the doped samples and to evaluate its distribution in the investigated areas, as well as to verify the absence of extraneous impurity contamination.

Thus, the applied set of experimental methods - diffusion doping, controlled α-irradiation, impedance spectroscopy, electron microscopy, and energy-dispersive analysis - provided a comprehensive investigation of the influence of rhodium and radiation exposure on the electrophysical properties of n-type silicon.

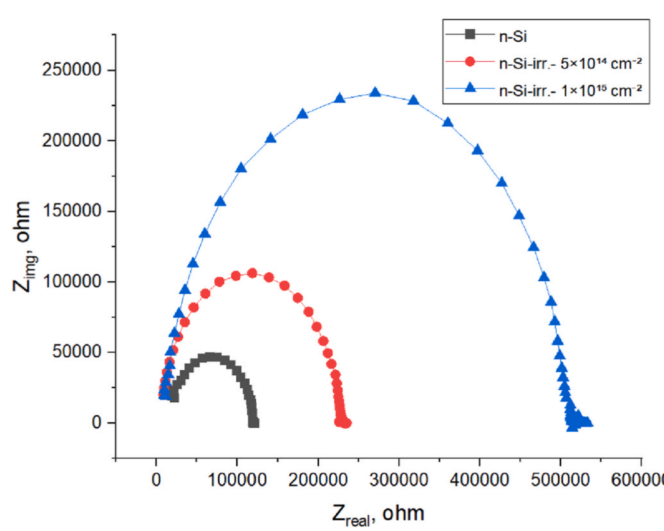


Fig. 1. Nyquist plots of the initial n-Si sample: before irradiation (black curve), after irradiation with a fluence of 5×10^{14} cm^{-2} (red curve), and 1×10^{15} cm^{-2} (blue curve). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3. Results and discussion

Fig. 1 shows the Nyquist plots for the initial and α-particle-irradiated n-type silicon samples. All impedance spectra exhibit a pronounced semicircular shape, indicating a dominant contribution of the bulk resistance and capacitance to the electrical response of the samples.

For the non-irradiated n-Si sample, the bulk resistance determined from the diameter of the semicircle is 127.7 kΩ. After irradiation with α-particles of 2 MeV energy, a significant increase in resistance is observed. At a fluence of 5×10^{14} cm^{-2}, the resistance increases to 224.6 kΩ, whereas at a fluence of 1×10^{15} cm^{-2} it reaches 526.2 kΩ, i.e., increases by more than a factor of four compared to the initial value.

Such a sharp increase in resistance indicates a substantial degradation of the electrical conductivity of silicon under α-irradiation. This behavior is associated with the formation of radiation-induced defects (vacancies, interstitial atoms, and their complexes), which introduce deep levels into the forbidden energy gap and efficiently trap charge carriers [Kozlov and Kozlovskiy, 2001]. As a result, the free carrier concentration decreases, leading to an increase in the bulk resistance. The systematic increase in the semicircle diameter in the Nyquist plots with increasing irradiation dose confirms the dominant role of bulk radiation-induced defects [Yaxia et al., 2025].

Fig. 2 shows the impedance spectra of n-Si samples doped by diffusion at 1100 °C before and after α-irradiation. In the initial state, the resistance of the doped sample is 85.6 kΩ, which is significantly lower than that of undoped n-Si (127.7 kΩ). This indicates an improvement in electrical conductivity due to the introduction of Rh, most likely as a result of modification of the defect structure and/or the formation of electrically active centers.

After irradiation with a fluence of 5×10^{14} cm^{-2}, the resistance increases to 192.2 kΩ, while at a fluence of 1×10^{15} cm^{-2} it reaches 416.1 kΩ. Despite the significant increase in resistance with increasing dose, its absolute values remain lower than those of undoped n-Si under the same irradiation conditions (224.6 kΩ and 526.2 kΩ, respectively).

The obtained results indicate that Rh doping at 1100 °C provides a certain improvement in the radiation hardness of the material. Most likely, Rh atoms participate in the formation of complexes with radiation-induced defects, partially compensating or passivating their electrical activity. However, at a high irradiation dose (1×10^{15} cm^{-2}), the resistance still increases by almost a factor of five (from 85.6 to 416.1 kΩ), which indicates a limited protective effect at this diffusion temperature [Hazdra and Komarnitskyy, 2009].

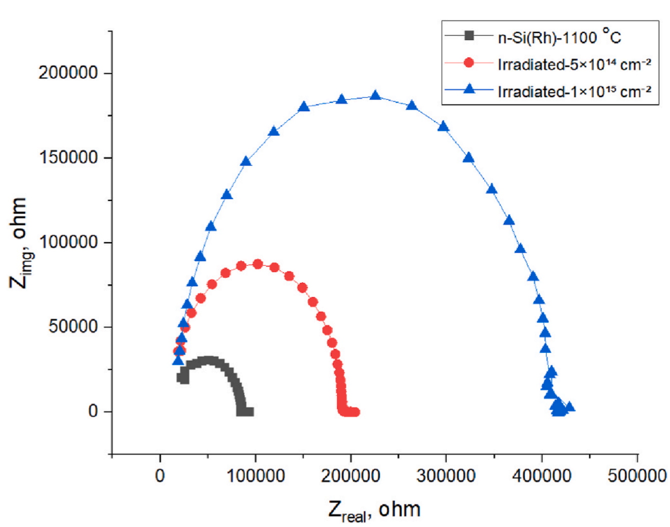


Fig. 2. Nyquist plots of n-Si samples doped with Rh at 1100 °C: before irradiation (black curve), after irradiation with a fluence of 5×10^{14} cm^{-2} (red curve), and 1×10^{15} cm^{-2} (blue curve). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Fig. 3 shows micrographs of the initial and doped silicon samples. In the images of rhodium-doped silicon (Fig. 3b and c), it is clearly seen that the diffusion doping process leads to the formation of various microstructures on the surface of the material. These microstructures may be associated with local modifications of the crystal lattice, as well as with the formation of dislocations and other defects arising from the incorporation of dopant atoms.

Additional experiments showed that rhodium doping of silicon at 1100 °C (Fig. 3b) is accompanied by the formation of structural microdefects with relatively low density. Increasing the doping temperature to 1200 °C (Fig. 3c) leads to the formation of smaller defects but at a higher concentration. This indicates that the diffusion kinetics and the interaction of rhodium atoms with the silicon crystal lattice strongly depend on the process temperature, which is reflected in the morphology and density of the microstructural defects in the material [Uzbas and Aydin, 2020; Utamuradova et al., 2024]. After α-particle irradiation, no significant changes in surface morphology were observed in the irradiated samples.

X-ray spectral analysis confirms the presence of rhodium in the silicon single crystals. According to the measurements, the rhodium content in the investigated samples is approximately 0.68 at.% (1.8 wt%). Energy-dispersive spectra (Fig. 4) also revealed the presence of additional elements, such as oxygen (0.95 at.% or 1.51 wt%) and carbon (1.12 at.% or 0.87 wt%). These results indicate that, in addition to the dopant rhodium atoms, impurities are present on the silicon surface, which may originate during the diffusion doping process or from the material's contact with the ambient atmosphere.

It should be noted that the variation in doping temperature had practically no effect on the concentration of rhodium atoms on the silicon surface, indicating a high stability of the dopant distribution in the surface layer. However, the rhodium concentration in the silicon bulk does depend on the diffusion temperature. These results emphasize the importance of controlling the temperature and diffusion conditions when modifying the properties of silicon with transition metals such as rhodium, as this directly affects the formation of microdefects and the structural characteristics of the material.

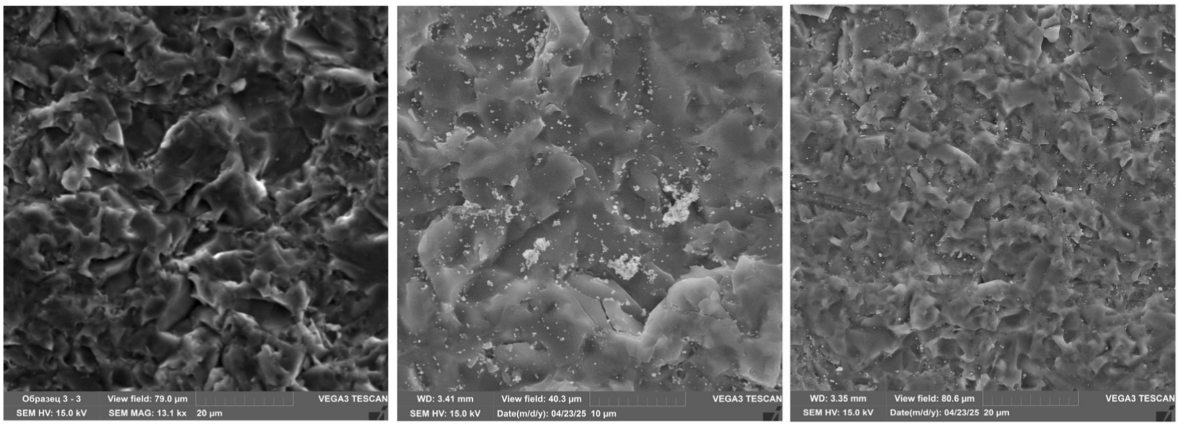
A significantly more pronounced effect is observed for samples doped with Rh at the higher diffusion temperature of 1200 °C (Fig. 5). In the initial state, these samples exhibit the lowest resistance among all investigated structures - 64.2 kΩ - indicating a substantial improvement in electrical properties due to high-temperature Rh diffusion.

After α-irradiation with a fluence of 5×10^{14} cm^{-2}, the resistance increases only to 104.3 kΩ, and even at the maximum fluence of 1×10^{15} cm^{-2}, it reaches just 148.2 kΩ. For comparison, at the same doses, the resistance of undoped n-Si is 526.2 kΩ, and that of samples doped at 1100 °C is 416.1 kΩ.

Thus, the relative increase in resistance for samples doped at 1200 °C is significantly smaller (from 64.2 to 148.2 kΩ, i.e., approximately 2.3 times) than in the other two cases. This clearly indicates a substantially higher radiation hardness of such structures.

This behavior can be explained by a more uniform distribution of Rh atoms and their more effective interaction with intrinsic and radiation-induced defects in silicon at the higher diffusion temperature. The formation of stable, electrically less active “dopant–defect” complexes suppresses the creation of deep levels and reduces the efficiency of charge carrier trapping, which allows the electrical conductivity to be largely preserved even after intense irradiation [Mccluskey and Haller, 2012; Pelenitsyn and Korotaev, 2022].

The formation of rhodium-defect complexes during high-temperature doping can be associated with the enhanced mobility of both Rh atoms and intrinsic point defects in silicon at elevated temperatures. Under such conditions, vacancies and self-interstitials become highly mobile and can effectively interact with diffusing Rh atoms, facilitating the formation of stable complexes such as Rh-vacancy and Rh-interstitial pairs. In addition, high-temperature processing



a *b* *c*

Fig. 3. Micrographs of the surfaces of single-crystal silicon wafers: (a) before doping, (b) after Rh doping at 1100 °C, and (c) after Rh doping at 1200 °C.

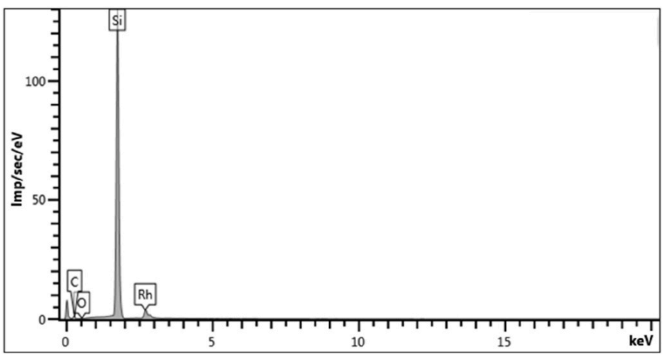


Fig. 4. Energy-dispersive spectra of single-crystal silicon doped with rhodium atoms.

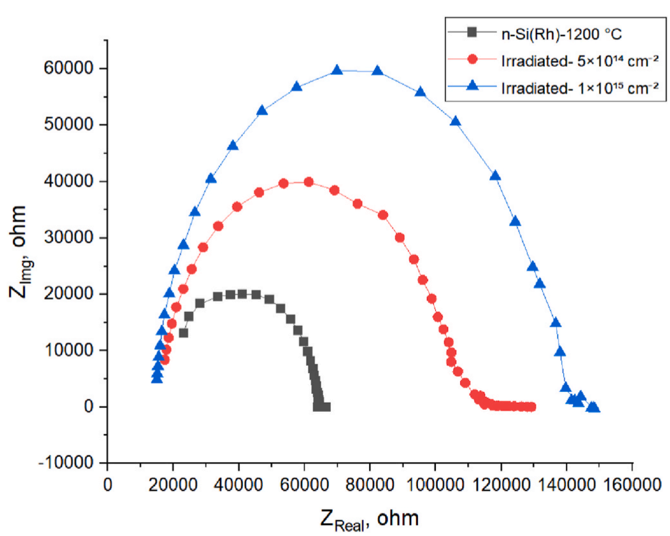


Fig. 5. Nyquist plots of n-Si samples doped with Rh at 1200 °C: before irradiation (black curve), after irradiation with a fluence of 5 × 10^{14} cm^{-2} (red curve), and 1 × 10^{15} cm^{-2} (blue curve). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

promotes the dissociation of pre-existing defect clusters and their reconfiguration into energetically favorable dopant-defect complexes. Upon subsequent cooling, these complexes can become electrically inactive or less active due to lattice relaxation and charge compensation effects. Furthermore, irradiation-induced defects introduced after doping can be efficiently captured by Rh atoms, acting as gettering centers, which further stabilizes the defect structure. As a result, the combined effect of thermally activated diffusion and defect interaction kinetics leads to the formation of a stable rhodium–defect subsystem that significantly influences the electrical properties of the material [Lindstrom, 2003; Schriefl et al., 2011].

A comparative analysis (Table 1) of all investigated samples demonstrates that both the α-irradiation fluence and the Rh diffusion temperature strongly influence the electrical properties of n-Si. Although irradiation in all cases leads to an increase in resistance due to the generation of radiation-induced defects, Rh doping significantly mitigates this effect, particularly at 1200 °C.

The deviation of the impedance spectra of n-Si⟨Rh⟩ samples doped at 1200 °C and subjected to α-irradiation from an ideal semicircular shape indicates a non-Debye type of relaxation (Fig. 6). This behavior reflects the presence of a distribution of relaxation times caused by the structural and defect inhomogeneity of the material [Asghar et al., 1994]. High-temperature Rh doping followed by irradiation leads to the formation of a complex defect–dopant subsystem, which includes various types of "dopant–defect" complexes and radiation-induced centers that contribute to both conductivity and polarization [Aguiar et al., 2025]. As a result, the electrical response of the samples cannot be adequately described by a simple RC model and requires the use of elements with distributed parameters, such as constant phase elements (CPE). The Nyquist plots of the investigated samples can be represented by a single equivalent electrical circuit (Fig. 6).

In addition, the deviation of the Nyquist plots from an ideal semicircle for samples doped at 1200 °C can be attributed to several interrelated factors. At such high doping temperatures, the diffusion of Rh atoms into the silicon lattice becomes significantly enhanced, leading not only to substitutional incorporation but also to the formation of clusters and complex defect configurations. This results in spatial fluctuations of electrical properties, such as local conductivity and

Table 1
Maximum values of ohmic resistance for Si samples doped with Rh before and after α-particle irradiation.

	Resistance [kohm]		
	n-Si	n-Si⟨Rh⟩ 1100 °C	n-Si⟨Rh⟩ 1200 °C
Before irradiation	127.7	85.6	64.2
After irradiation with dose 5 × 10^{14} cm^{-2}	224.6	192.2	104.3
After irradiation with dose 1 × 10^{15} cm^{-2}	526.2	416.1	148.2

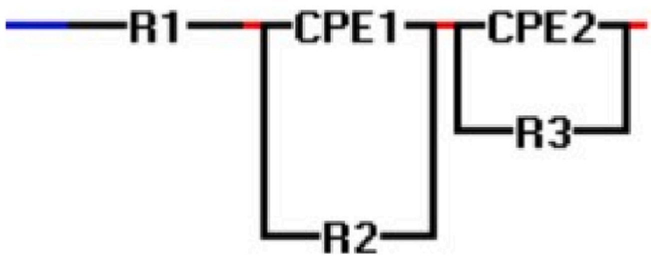


Fig. 6. Equivalent electrical circuit, where R represents the resistive element and CPE denotes a constant phase element.

permittivity. Furthermore, α-irradiation introduces additional point defects and defect clusters, which interact with Rh atoms and create electrically active centers with different relaxation characteristics. These processes give rise to a superposition of multiple relaxation mechanisms rather than a single relaxation time. Consequently, the impedance response becomes dispersive, and the Nyquist diagram deviates from the ideal Debye-type semicircle, reflecting the non-uniform and heterogeneous nature of the defect–dopant system in the material [Donegani et al., 2018; Utamuradova et al., 2026].

The exposure of silicon to ionizing radiation leads to the formation of vacancies, interstitial atoms, and their complexes, which create deep levels in the forbidden energy gap and degrade the electrical properties of the material [Messenger and Ash, 1992; Pintilie et al., 2009; Alyamani and Mustapha, 2016]. Numerous studies have shown that such radiation-induced defects efficiently trap charge carriers and act as recombination centers, resulting in a decrease in free carrier concentration and an increase in both specific and bulk resistivity [Pintilie et al., 2009; Alyamani and Mustapha, 2016].

The results obtained in the present work for undoped n-Si are fully consistent with these observations. The observed sharp increase in resistance after α-irradiation is attributed to the accumulation of radiation defects analogous to so-called A-centers and more complex defect complexes, widely discussed in the literature on radiation damage in silicon [Watkins, 2000].

One effective approach to enhancing the radiation hardness of silicon is intentional doping with impurities capable of interacting with radiation-induced defects and modifying their electrical activity [Markevich et al., 2009]. Several studies have shown that transition metal impurities (Pt, Cr, Gd, etc.) can form complexes with vacancies and interstitial atoms, thereby reducing the concentration of electrically active radiation centers and mitigating the degradation of electrophysical properties after irradiation [Ceponis et al., 2020; Xian et al., 2025].

The results of the present study for n-Si⟨Rh⟩ samples doped at 1100 °C and 1200 °C fit well into this concept. In particular, it is noteworthy that at the higher doping temperature of 1200 °C, the increase in resistance after irradiation is significantly smaller than that of undoped samples and those doped at 1100 °C. This indicates more effective formation of stable "dopant–defect" complexes during high-temperature diffusion, analogous to effects reported for other systems such as Si⟨Pt⟩, Si⟨Pd⟩, and related materials [Utamuradova et al., 2023b; Utamuradova et al., 2025c].

4. Conclusion

In this work, the effect of α-irradiation with an energy of 2 MeV and fluences of 5×10^{14} and 1×10^{15} cm^{-2} on the electrophysical properties of n-type silicon was systematically investigated using impedance spectroscopy. The study also examined the role of rhodium doping performed at 1100 °C and 1200 °C in enhancing the radiation hardness of the material.

It was shown that irradiation of undoped n-Si leads to a sharp deterioration of its electrical characteristics. The bulk resistance of the initial sample increases from 127.7 kΩ to 224.6 kΩ at a fluence of 5×10^{14} cm^{-2} and to 526.2 kΩ at a fluence of 1×10^{15} cm^{-2}, indicating intensive accumulation of radiation-induced defects that form deep levels and effectively reduce the concentration of free charge carriers.

Rhodium doping at 1100 °C reduces the initial resistance to 85.6 kΩ and somewhat mitigates the adverse effects of α-irradiation. However, at the high fluence of 1×10^{15} cm^{-2}, the resistance still increases to 416.1 kΩ, indicating the limited effectiveness of this doping regime under intense radiation exposure.

The most significant enhancement of radiation hardness was observed for samples doped with rhodium at 1200 °C. In these samples, the initial resistance is 64.2 kΩ and, after irradiation, increases only to 104.3 kΩ (5×10^{14} cm^{-2}) and 148.2 kΩ (1×10^{15} cm^{-2}), representing more than a twofold, but substantially smaller, change in electrical parameters compared to the other samples.

Comparative analysis shows that, at the maximum fluence of 1×10^{15} cm^{-2}, the resistance decreases in the sequence: n-Si (526.2 kΩ) → n-Si⟨Rh⟩-1100 °C (416.1 kΩ) → n-Si⟨Rh⟩-1200 °C (148.2 kΩ), unequivocally confirming the critical role of high-temperature rhodium diffusion doping in forming a radiation-resistant defect–dopant structure in silicon.

These results allow us to conclude that rhodium, particularly when introduced at high temperatures, promotes the formation of stable complexes with intrinsic and radiation-induced defects in silicon, reducing their electrical activity and thereby significantly mitigating the degradation of electrophysical properties under α-irradiation.

From a practical standpoint, the obtained results are of considerable importance for the development of radiation-resistant micro- and nanoelectronic devices intended for operation in harsh environments, such as space systems, nuclear energy, and high-energy physics installations. The demonstrated ability of high-temperature rhodium doping to significantly suppress radiation-induced degradation of electrical parameters suggests a viable technological approach for enhancing the operational stability and lifetime of silicon-based components. In particular, such materials can be effectively utilized in sensors, power electronics, and integrated circuits where reliability under high radiation doses is a critical requirement. Thus, the implementation of rhodium doping at elevated temperatures can be considered a promising strategy for improving the radiation tolerance of semiconductor devices in advanced technological applications.

CRediT authorship contribution statement

Sh.B. Utamuradova: Supervision, Conceptualization. **D.A. Rakhmanov:** Writing – original draft, Project administration, Formal analysis. **A.S. Doroshkevich:** Resources, Data curation. **Zh.V. Mezentseva:** Visualization, Methodology. **R.Sh. Isayev:** Validation, Funding acquisition. **A. Tatarinova:** Writing – review & editing, Software.

Declaration of competing interest

The authors declare that they have no known competing financial or personal interests that could have appeared to influence the work reported in this paper. The authors confirm that this research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. No funding sources, institutions, or organizations had any role in the study design, data collection, analysis, interpretation of the results, or in the decision to publish the manuscript. The authors also declare that they have no professional, academic, or other relationships that could be perceived as influencing the objectivity, integrity, or impartiality of the presented results.

Data availability

Data will be made available on request.

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